SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 14

ONLINE EXAMINATION SYSTEM – NEGATIVE TEST SCENARIOS AND TEST CASES

Aim:

To Write a case study to identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases for the application.

1. Problem Description

An online examination system allows registered students to authenticate, view assigned examinations, read instructions, answer questions, save responses and submit within the permitted examination window.

The Online Examination System is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Only registered students may access their assigned examination.
2. The examination must be available only during the configured time window.
3. Instructions must be displayed before the examination starts.
4. The system must save selected answers reliably.
5. The timer must reflect the configured examination duration.
6. Students must not submit after the permitted deadline unless an extension is authorized.
7. Submission must be recorded once and produce a confirmation.
8. Results must be visible only when released by an authorized role.

4. Negative Test Scenarios and Test Cases

Negative testing checks whether invalid, incomplete, unauthorized or boundary inputs are safely rejected with a clear message.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Login with invalid credentials	Enter incorrect student password	Login is denied.	Pass
NT02	Open unassigned exam	Attempt to access exam not assigned to student	Access is denied.	Pass
NT03	Start before window	Open exam before configured start time	Exam cannot be started.	Pass
NT04	Start after deadline	Attempt to start after exam closes	Access is blocked unless extension exists.	Pass
NT05	Submit after expiry	Attempt normal submission after deadline	System applies configured deadline policy and prevents unauthorized late submission.	Pass
NT06	Save invalid option	Submit an option not belonging to the question	Invalid answer is rejected.	Pass
NT07	Submit repeatedly	Click submit multiple times	Only one final submission is recorded.	Pass
NT08	View unreleased result	Open result before authorized release	Result remains hidden.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: NT05

Precondition: The application is running in the test environment and the relevant test data is available.

Test Steps: 1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Attempt normal submission after deadline. 4. Observe the application response.

Expected Result: System applies configured deadline policy and prevents unauthorized late submission.

Actual Result: For demonstration, assume the application accepts the invalid input or does not display the required validation message.

Conclusion: The test is marked as failed for defect-report demonstration. This is an illustrative result, not a claim about a deployed application.

6. Sample Bug Report

Field	Description
Bug ID	BR-014
Linked Test Case	NT05
Severity	High (illustrative)
Priority	High (illustrative)
Bug Title	Online Examination System does not correctly validate the condition: Submit after expiry.
Environment	Test environment; browser and operating system should be recorded during actual execution.
Preconditions	The application is running in the test environment and the relevant test data is available.
Steps to Reproduce	1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Attempt normal submission after deadline. 4. Observe the application response.
Expected Result	System applies configured deadline policy and prevents unauthorized late submission.
Actual Result	For demonstration, assume the application accepts the invalid input or does not display the required validation message.
Impact	Invalid data or an unauthorized operation may be accepted, resulting in incorrect processing, data integrity issues or a poor user experience.
Suggested Fix	Apply validation at both client and server sides, return a clear error message, and add regression tests for the boundary and invalid condition.
Status	Open – illustrative sample.

7. Bug Life Cycle

New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still reproducible after the correction, the tester reopens it and returns it to the development team.

Conclusion

The test cases prepared for the Online Examination System provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.